[INCH-POUND]

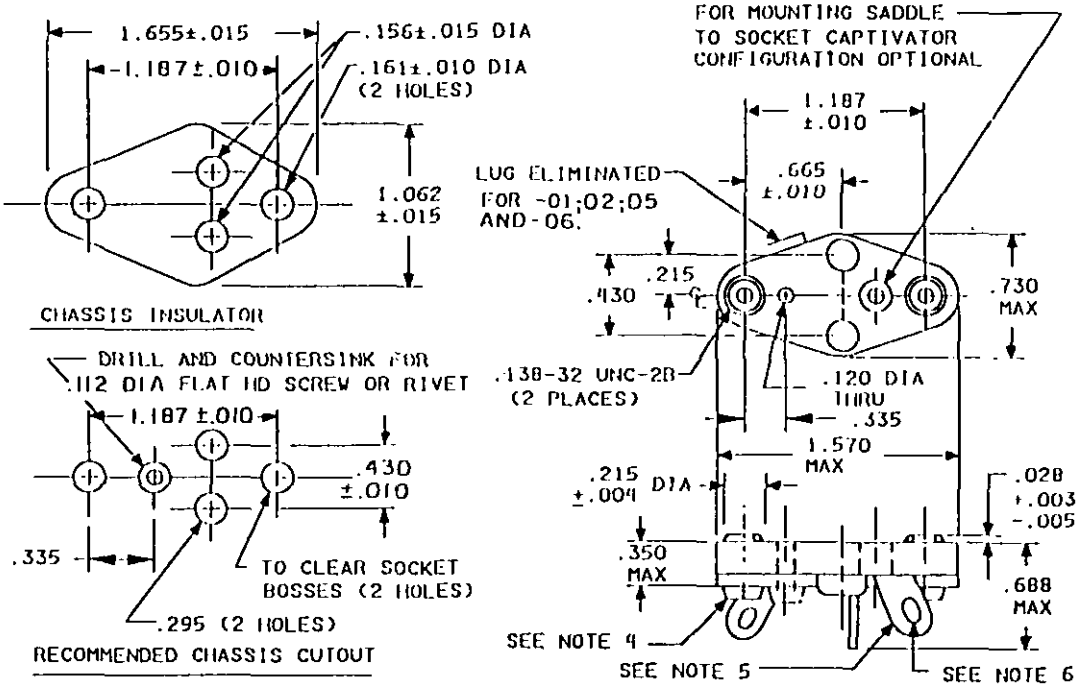
MIL-S-12883/42E
19 September 1989
SUPERSEDING
MIL-S-12883/42D
22 May 1987

MILITARY SPECIFICATION SHEET

SOCKETS AND ACCESSORIES FOR PLUG-IN ELECTRONIC COMPONENTS (POWER TRANSISTOR, 2 CONTACT, 10 AMPERES, TO-3)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-12883



Inches	mm
.003	0.08
.004	0.10
.005	0.13
.010	0.25
.015	0.38
.028	0.71
.112	2.84
.120	3.05
.138	3.51
.156	3.96
.161	4.09
.215	5.46
.295	7.49
.335	8.51
.350	8.89
.430	10.92
.665	16.89
.688	17.48
.730	18.54
1.062	26.97
1.187	30.15
1.570	39.88
1.655	42.04

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005.
4. Direction of threaded boss optional, may face inward or outward.
5. Lug shall be bent between 45 and 90 degrees and must remain in the confines of the .730 max and 1.570 max dimensions. Location of lug optional.
6. Lug and contact tabs shall have either 1 wire hole of .040 (1.02 mm) minimum width and .120 (3.18 mm) minimum length, or 2 holes of .040 (1.02 mm) minimum width and .075 (1.91 mm) minimum length. The hole, or holes shall lie on the longitudinal centerline of the contact tab within ±.008 (0.20 mm).

FIGURE 1. Socket configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Material:

Insulator: Diallyl phthalate, green, in accordance with MIL-M-14, type SDG-F.

Contacts: Beryllium copper per QQ-C-533 or Spinodal alloy C72900 (Pfinodal).

Contact finish: See table I.

Mounting saddle and eyelet (use of eyelet optional): Brass in accordance with QQ-B-626 or QQ-B-613, plated tin-lead, 50 to 70 percent tin, in accordance with MIL-P-81728.

Chassis insulator: 0.003 inch (0.08 mm) thick mica. For replacement purposes, order M38527/8-05P in accordance with MIL-M-38527.

Minimum thread strength (torque):
18 inch-pounds for dash numbers 01 through 04.
12 inch-pounds for dash numbers 05 through 08.

Contact clearance: Clearance between the cavity wall and the contact, with the contact in any position, shall be such that a 0.038 inch (0.97 mm) round nose pin with a force of 5 pounds will not penetrate between the cavity and contact.

Application: Sockets to accept components with 0.040 +.003, -.002 inch (1.02 +0.08, -0.05 mm) diameter leads.

Float: With a pin of 0.043 inch (1.09 mm) diameter and 0.271 inch (6.88 mm) length fully inserted in a socket contact, the contact shall be capable of free movement (float) within the contact cavity.

Insertion and withdrawal force: On the insertion and withdrawal of the mating component and socket, the insertion and withdrawal forces of the socket and mating component shall be as follows:

Insertion force: 6.00 pounds, maximum.
Withdrawal force: 0.18 pound, minimum; 6.00 pounds, maximum.

Insulation resistance: 1,000 megohms, minimum. Test pin diameter, size 16, 0.040 +.003, -.002 inch.

Dielectric withstanding voltage:

Sea level: Test voltage, 1,500 V rms. Test pin diameter, size 16, 0.040 +.003, -.002 inch.

High altitude (80,000 feet): Test voltage, 500 V rms. Test pin diameter, size 16, 0.040 +.003, -.002 inch.

Vibration: Sockets shall meet the requirements of method 204, test condition B, MIL-STD-202 (10-2,000 Hz) with a power transistor inserted for use as a test gauge.

Contact resistance:

Average for all contacts: 0.015 ohm, maximum.
Individual contacts: 0.03 ohm, maximum.

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Mechanical shock: Sockets shall meet the shock requirements specified in MIL-S-12883 with a power transistor inserted for use as a test gauge.

Durability (50 cycles): After the durability test, the insertion force and withdrawal force shall be 6 pounds, maximum and .125 pound, minimum.

Weight (maximum): 2 ounces.

Test gauge details: See table II and MIL-S-12883.

Marking: Sockets shall be marked on the side or the top surface.

Part or Identifying Number: M12883/42-(and dash number from table I).

TABLE I. Dash numbers, contact and termination finish, and mounting thread sizes.

Dash number M12883/42	Contact engagement area and termination finish	Mounting thread sizes
-01 (1 lug)	Gold - MIL-G-45204, type II, class 0 (0.000030 inch) over nickel (0.000050 to 0.000100 inch)	.138-32 UNC-2B
-02 (1 lug)	Tin-lead plating, MIL-P-81728 50 to 95% tin (.000100 to .000300 inch)	.138-32 UNC-2B
-03 (2 lugs)	Gold MIL-G-45204, type II, class 0 (0.000030 inch) over nickel (0.000050 to 0.000100 inch)	.138-32 UNC-2B
-04 (2 lugs)	Tin-lead plating, MIL-P-81728 50 to 95% tin (.000100 to .000300 inch)	.138-32 UNC-2B
-05 (1 lug)	Gold - MIL-G-45204, type II, class 0 (0.000030 inch) over nickel (0.000050 to 0.000100 inch)	M3X0.5
-06 (1 lug)	Tin-lead plating, MIL-P-81728 50 to 95% tin (.000100 to .000300 inch)	M3X0.5
-07 (2 lugs)	Gold MIL-G-45204, type II, class 0 (0.000030 inch) over nickel (0.000050 to 0.000100 inch)	M3X0.5
-08 (2 lugs)	Tin-lead plating, MIL-P-81728 50 to 95% tin (.000100 to .000300 inch)	M3X0.5

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TABLE II. Test gauge details.

Inspection	A (Test pin length)	B (Test pin dia)	M (Test-end diameter)	J (Test pin center to center)	N (Probe-end diameter)	Total weight of gauge (±5 percent)
	Inch	Inch	Inch	Inch	Inch	Ounces
Insertion and withdrawal force 1/	.150 min	0.0380 ±0.0001		.4300 basic		
Contact resistance 2/	.150 min	0.0380 ±0.0001		.4300 basic		
Contact retention 3/	.150 min		0.0380 ±0.0001		0.0430 ±0.0001	2
Durability 1/	.150 min	0.0430 ±0.0001		.4300 basic		

1/ Test gauge shall be similar in construction to figure 1C of MIL-S-12883.

2/ Contact resistance shall be measured using a brass jumper having pin length, diameter and center-to-center spacing as shown. The test leads shall be attached to the terminals as close to the socket body as possible.

3/ See figure 4 of MIL-S-12883.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ER, ME
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3682-06)

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